

Deep Watch

Reasoning Under Pressure Beneath the Surface

15 WATCHES · 61 STOPS · GRADE 12 · DEEPWATCH

THE OPENING CARD

You have the watch on a submarine at ninety metres. Everybody aboard is asleep, on watch, or working in a compartment they cannot see out of. Neither can you. What you know about the water comes from sonar. What you know about the boat comes from gauges other people read out to you over a telephone. A contact nobody has identified is closing, and water already inside the hull is rising. The pencilled track on the chart drifts further from where the boat really is with every minute nobody corrects it. Surfacing is where every one of these problems stops being fatal. Somebody has to be willing to call it.

Boat Walkdown

BEFORE THE WATCH — GREET

Nobody sails with people they have not met

You joined at the last port and the boat sailed the same night. Rask, the sonar supervisor, wants you known to every watch before the first drill, because in a compartment with the lights out a voice you recognise is worth more than a nameplate you cannot read. Get through the boat and put a name to most of the crew before the watch changes.

PLAN CARD 102W

Two hours to sailing, alongside, and the fire main isolation valve in the forward compartment is chained shut with nobody willing to say who chained it. Chief Ferro wants to walk you through the boat before you take your first watch tonight. This watch you learn the forward spaces, what the fittings on a bulkhead are for, and how to get from one end of the boat to the other without looking at anything. Ferro's rule is that a route learned as a sequence survives being interrupted. Tonight you may have to find that valve in the dark, in smoke, by feel.

BRIEFING 42W

You have the watch for the first time. Before anything else happens, you have to be able to find your way around this hull in the dark, and know what is in each space and how you would get out of it.

WHAT YOU DECIDE 13W

Learn the boat as a system of spaces, not a list of names.

1.1 What a walkdown is for

DAMAGE CONTROL · FORWARD EQUIPMENT & HANDLING · A ROOM · SEQUENCE

SCENE 42W

Chief Ferro walks you into the forward space and asks you to learn it. Equipment down both sides, deck plates over a recess, a hose reel on the bulkhead, two ways out. He is not telling you what to take in first.

GUIDE 67W

All four of these can be done in the same minute, so this is not a timetable. Ask instead what each fact lets you interpret. A locker of shoring means nothing until you know what this space can do to you. A hatch you have not tried is an assumption. At ninety metres, one of these decides whether a casualty is this compartment's or the next one's.

ASK 12W

Learn a compartment in the order the facts depend on each other

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a submarine is divided into compartments that can be sealed off

VERDICT 80W

A submarine survives a casualty by being divided. The pressure boundary comes first: at 90 metres it separates a flooded compartment from a flooded boat. The ways out come next, and they have to be proved rather than assumed. Then find what could start a casualty here: sea connections, cables or stores. Last, judge whether the equipment stored here can fight those threats. A locker is only the right locker once you know what this space can do to you.

TAKEAWAY 15W

A compartment is a pressure boundary first; everything you learn about it hangs off that.

1.2 Every fitting is an argument

PROPULSION & ENGINEERING · PROPULSION MACHINERY · A PERSON · PROTOCOL

SCENE 39W

Chief Haruki stops beside four fittings on the same bulkhead and asks what each one is for. None carries a purpose label, only a number. He steps back from the valves and waits for you to make the distinctions.

GUIDE 67W

None of these four carries a purpose label, only a number. So work from shape, height and what lies beyond. Ask of each what is on the far side of it, and what opening it would admit or supply. The consequence is the clue. At ninety metres one of these has the sea behind it at ten atmospheres, and the wrong wheel is not a small mistake.

ASK 10W

Match each fitting to what it is there to do.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

pressure rises about one atmosphere for every ten metres of depth · pressure, depth and hydrostatics — taken as read

VERDICT 80W

A fitting makes sense only when you connect its shape to what lies beyond it. A wheel valve through the pressure hull may have the sea behind it at high pressure. A quick-connect manifold at head height can supply breathing air. A strapped locker can hold shoring and plugs. A hose reel can supply firefighting water. The point is not to memorize four silhouettes. It is to infer function from connection, location and the consequence of operating the wrong thing.

TAKEAWAY 20W

What a fitting is for is decided by what is on the other side of it, and at what pressure.

1.3 The trip back without a waypoint

NAVIGATION & POSITION · CONTROL ROOM · A ROOM · ROUTE

SCENE 39W

The navigator sends you from control to machinery with a sealed note. On the return trip one watertight door will be shut and the marker lights will be off. You have walked the route once, with labels still visible.

GUIDE 66W

You walk the route once with the labels showing. Then the labels go and you run it back from memory. One watertight door will be shut and the marker lights will be off. When that happens, find the landmark in the compartment you are actually in and work out where the detour has put you. You are graded on the compartments you reach, not the time.

ASK 22W

Run the route from memory. After the blocked door, use the next landmark to work out where the detour has put you.

BEHIND THE BUTTON 136W

Why places beat counting. A remembered count of turns survives only while nothing interrupts it. Miss one and every number after it is wrong, with nothing to tell you so. A place is different: two helm wheels means Control whether you arrived as planned or by a detour.

What the landmarks are for. Every compartment holds something that is nowhere else on the boat — the wheels, the radio racks, the long mess table, a particular pump. That is how a crew navigates in smoke or darkness, and it is what the second run is testing.

Why the door is shut without warning. Routes are learned in good conditions and used in bad ones. A route that only works when every door is open is not a route, it is a memory of one particular walk.

TAKES AS READ

a submarine is divided into named compartments · a route can be remembered as a sequence of places

VERDICT 73W

A count only works while nothing interrupts it. A turn list fails after one extra turn. A route made of places is different because each compartment becomes a new reference point. If you are diverted, identify the space you enter. Then rejoin the sequence from there. Submarines use the same logic when dead reckoning. The useful question is not how many steps have passed. It is what trusted reference remains after the interruption.

TAKEAWAY 18W

A route remembered as places can be recovered after an interruption; a count or turn list usually cannot.

A compartment is defined by its boundaries, what it holds, and how you leave it.

Contact in the Noise

BEFORE THE WATCH — TRIAL

Learn the boat before you need to move through it in the dark

One passage, nine compartments, and every bulkhead door a hatch you have to fold yourself through. Walk the whole length now, while the lights are on and nothing is wrong, because the next time you do this at speed it will be to a compartment that is filling and you will not be able to see the deck.

PLAN CARD 105W

Second day out, forenoon watch. 4 traces on the broadband display, and the sonar supervisor, Nils Rask, has been arguing with his own operator about the fourth for 11 minutes: it holds almost the same bearing and it is getting louder. The captain, Commander Iris Vance, wants a classification before the boat commits to the track she has been ordered onto. This watch you sort out what is out there and say what a steady bearing means. Sound is all you have. And a bearing that does not change is not a quiet situation — it is the signature of two ships on converging courses.

BRIEFING 43W

Sonar has four sources on the display at once. Two of them have been there since before the boat slowed, one came with the turn, and one comes and goes. The captain wants to know which of them she has to care about.

WHAT YOU DECIDE 12W

Work out what is out there before the boat commits to anything.

2.1 Four traces, one of them ours

SONAR & ACOUSTICS · SONAR ROOM · A ROOM · PROTOCOL

SCENE 42W

Four traces sit on the broadband display while the watch calls them out. The boat slowed four minutes ago, and one trace changed with it. The others kept their own patterns. Rask wants each source identified before the next sweep rolls across.

GUIDE 59W

Each trace carries more than one clue. How steady is the bearing? How wide is the spectrum? How long does it last? And did it follow the boat's own change of speed? Pair on the combination, not on which trace is loudest. The boat slowed four minutes ago, and that is a test one of these has already taken.

ASK 8W

Match each trace to what is making it.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

passive sonar listens and does not transmit · sound as a wave: frequency, wavelength, speed — taken as read

VERDICT 82W

Each trace has more than one clue. A sound that follows own-ship speed changes is probably generated aboard. A narrow line that drifts slowly in bearing fits steady machinery on another vessel. Short chirps that vanish quickly fit a biologic source better than a hull. A nearly steady bearing that grows louder deserves attention because its geometry is changing differently from the others. Classification comes from combining time, bearing, spectrum and response to own-ship changes, not from picking whichever trace looks strongest.

TAKEAWAY 12W

A source that changes when the boat changes belongs to the boat.

2.2 The trace that will not resolve

SONAR & ACOUSTICS · SONAR ROOM · A PERSON · CHOICE

SCENE 33W

The fourth trace is faint, has held nearly the same bearing for 11 minutes, and does not appear on narrowband at all. The boat has not changed speed or depth since it appeared.

GUIDE 63W

Two facts have to be explained together: the trace is faint, and it has no narrowband line. Ask of each option whether it accounts for both. A layer can bend a path and make a source weak. It does not strip machinery tones away selectively. Eleven minutes is a long life for a processor artefact, and nothing aboard changed when the trace appeared.

ASK 10W

Which explanation fits every reading, not just the loudest one?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

sound bends when it crosses water of a different temperature

VERDICT 76W

Broadband measures sound energy across a band. Narrowband looks for stable tonal lines. A sound-speed layer can bend a path and make an external source weak, but it does not selectively strip away machinery tones. Faintness and missing tonals are therefore separate clues. The layer can explain the weak level; the absent narrowband says no tonal line is above threshold. 11 minutes of persistence argues against a transient artefact, and nothing aboard changed with the trace.

TAKEAWAY 16W

Refraction can weaken a received path, but it does not selectively erase the source's spectral content.

2.3 The bearing that does not move

NAVIGATION & POSITION • CONTROL ROOM • A ROOM • CHOICE

SCENE 42W

The contact has stayed within half a degree of the same bearing for six minutes. Its level rises on every sweep. The boat is still making eight knots, and the officer of the deck wants an answer before the channel narrows ahead.

GUIDE 69W

A bearing is a direction and not a range, so no option here can be settled by bearing alone. Ask what each one would require you to already know. Steady bearing with a rising level does not prove a collision. What it does is remove the comfortable reading, that the contact is sliding safely across the bow. The channel narrows ahead and the boat is still making eight knots.

ASK 11W

What does a steady bearing and a rising level tell you?

BEHIND THE BUTTON 195W

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TAKES AS READ

passive sonar gives a direction and no range at all • sound as a wave: frequency, wavelength, speed — taken as read

VERDICT 82W

A bearing is the direction to a source, not its range. If the bearing stays nearly fixed while the sound grows stronger, the relative geometry is not opening sideways very quickly. That can happen when two tracks are converging. The important point is not that a steady bearing proves collision by itself. It is that steady bearing plus rising level removes one comforting explanation: the contact is not simply sliding safely across the bow. The watch needs range or motion information next.

TAKEAWAY 15W

A constant bearing with a rising level is a collision geometry, not a quiet one.

END OF THE WATCH 15W

A bearing is a measurement; a track is an argument built from several of them.

Position Without a Trusted Fix

BEFORE THE WATCH — FOLLOW

The rounds, walked once by the man who has done them for eleven years

Petrov, the assistant navigator, does engineering rounds twice a watch and has never written them down. Stay with him. He stops at each machine long enough to listen to it, which is the part that cannot be written down, and he does not slow up for anybody. Get too close and you are standing between him and the thing he is listening to.

PLAN CARD 121W

Third day, middle watch, and the last trusted fix was three and a half hours ago. The navigator, Imani Sowande, has the plot built from course, speed and time, her assistant Anya Petrov has been reading the depth sounder, and the two of them do not agree. There is a current here of about half a knot that nobody has measured today. This watch you work out how far the water has carried the boat, decide which instrument to believe, and get a position that does not depend on the plot at all. Two displays taking their heading from the same gyro will agree beautifully, which is Sowande's whole worry. 60 fathoms, and a bank half a mile off the track.

BRIEFING 42W

The boat has been submerged for hours on dead reckoning. The plot is a confident-looking line drawn from course and speed, and it knows nothing about the water that has been setting the boat sideways the whole time. Ahead is a bank.

WHAT YOU DECIDE 13W

Work out not where the boat is, but how well that is known.

3.1 How far the water has carried us

NAVIGATION & POSITION · CONTROL ROOM · A ROOM · BALLPARK

SCENE 35W

The last trusted fix was three and a half hours ago. The forecast set for this area is about half a knot, and nobody has applied it to the plot because nobody has measured it.

GUIDE 64W

Five numbers, and three belong to the boat rather than the water: own speed, depth and course. Ask of each whether it moves the boat through the water or carries the water itself. Speed through the water cannot see a current. Half a knot for three and a half hours is nearly two miles. This is the plot the boat enters the narrows on.

ASK 14W

Estimate how far the boat could be from where the plot says it is.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

the log measures speed through the water, not over the ground

VERDICT 78W

Between fixes, the boat carries its position forward from the last trusted point using course, speed and time. That is dead reckoning. The water moves too, and a current can carry the whole boat sideways without appearing in speed through the water. Half a knot for three and a half hours is nearly two miles of set. The error grows every hour it goes unmeasured. This is the same plot the boat will use to enter the narrows.

TAKEAWAY 12W

Dead reckoning tells you where you would be if nothing pushed you.

3.2 Two displays, one source

NAVIGATION & POSITION • CONTROL ROOM • A PERSON • CHOICE

SCENE 42W

The chart overlay and electronic repeat agree to within 100 yards. The depth sounder returns an echo after 0.120 seconds; sound speed is 1500 metres per second. The chart says 102 metres here. The last trusted position fix was 3.5 hours ago.

GUIDE 64W

Two of the readings here come from one source shown twice, so their agreement is not a second opinion. Sort the evidence by how many independent measurements stand behind it. The sounder is the only genuinely separate one. Work out what it says before choosing. A disagreement between an independent measurement and a repeated one tells you which needs checking, not which is wrong.

ASK 12W

What can you conclude from these readings before entering the banked water?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

an echo time is a round trip • two displays that share one source can share one error

VERDICT 83W

The sounder depth is half the round-trip distance: $\frac{1}{2} \times 1500 \text{ m/s} \times 0.120 \text{ s} = 90 \text{ m}$. The chart says 102 m, so something is wrong by 12 m. The two navigation displays do not settle it because both repeat the same inertial source. Their agreement is one measurement shown twice. The sounder is independent, so the disagreement is real evidence. It does not by itself prove whether the chart or position is wrong; it proves the position needs an independent check.

TAKEAWAY 12W

Agreement only counts as confirmation when the second measurement can fail differently.

3.3 Where the sound will not go

SONAR & ACOUSTICS · SONAR ROOM · A ROOM · CHOICE

SCENE 39W

The sound-speed trace reads 1505 metres per second above 60 metres and 1485 below it. A contact held for 40 minutes while this boat stayed at 20 metres, then vanished in 1 sweep while every other trace remained unchanged.

GUIDE 63W

The contact vanished in one sweep while every other trace held. Ask of each option whether it explains that suddenness. Range and shutdown both take effect gradually. The baffles are a direction rather than an event. The sound-speed numbers are what do the work here. Rays bend toward slower water, and a region can be left that rays from this depth never reach.

ASK 16W

The contact was held on the surface layer and is now lost. Where has it gone?

BEHIND THE BUTTON 195W

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TAKES AS READ

Snell's-law angles are measured from the normal to the layer · a lower sound speed bends a ray toward the normal

VERDICT 74W

At a layer, Snell's law gives $\sin\theta_1/v_1 = \sin\theta_2/v_2$. For a ray at 60° from the vertical, $\sin\theta_2 = (1485/1505)\sin 60^\circ$, so θ_2 is about 58.7° . The smaller angle means the ray bends more toward the vertical as it enters slower water. Across a real sound-speed profile, repeated bending can leave a region that rays from this depth do not reach. A contact entering that shadow can disappear abruptly instead of fading smoothly with range.

TAKEAWAY 15W

Sound bends toward slower water, which leaves regions a sonar cannot hear into at all.

3.4 What a walkdown is for

DAMAGE CONTROL · FORWARD EQUIPMENT & HANDLING · A ROOM · CALLBACK · SEQUENCE

SCENE 42W

Chief Ferro walks you into the forward space and asks you to learn it. Equipment down both sides, deck plates over a recess, a hose reel on the bulkhead, two ways out. He is not telling you what to take in first.

GUIDE 67W

All four of these can be done in the same minute, so this is not a timetable. Ask instead what each fact lets you interpret. A locker of shoring means nothing until you know what this space can do to you. A hatch you have not tried is an assumption. At ninety metres, one of these decides whether a casualty is this compartment's or the next one's.

ASK 12W

Learn a compartment in the order the facts depend on each other

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a submarine is divided into compartments that can be sealed off

VERDICT 80W

A submarine survives a casualty by being divided. The pressure boundary comes first: at 90 metres it separates a flooded compartment from a flooded boat. The ways out come next, and they have to be proved rather than assumed. Then find what could start a casualty here: sea connections, cables or stores. Last, judge whether the equipment stored here can fight those threats. A locker is only the right locker once you know what this space can do to you.

TAKEAWAY 15W

A compartment is a pressure boundary first; everything you learn about it hangs off that.

Precision is not accuracy, and two displays sharing a source are one measurement.

Silent Passage

PLAN CARD 110W

Fifth day, first watch. The narrows are 4 miles across, the transit takes 11 hours, and the plot's uncertainty is a circle 2 miles wide. The executive officer, Theo Whitfield, wants to wait for a better fix. Vance points out that waiting puts the boat in the middle of the strait at first light, which is worse than any argument about the plot — and this time she is right. This watch you decide what has to be true before the boat commits, how much margin the passage really has, and what Haruki can afford to shut down for Rask. What matters is how much water stays under the keel.

BRIEFING 30W

The route ahead narrows between the bank and a shipping lane. Command wants the boat through it with contact held throughout and as little acoustic exposure as the plant allows.

WHAT YOU DECIDE 12W

Cross a constrained area without losing contact awareness or making unnecessary noise.

4.1 What has to be true before the boat commits

SONAR & ACOUSTICS · SONAR ROOM · A ROOM · SEQUENCE

SCENE 38W

Rask closes the sonar-room door and lays out the passage checks. The narrows are already on the next chart panel. Once the boat commits, the channel leaves little room to turn around and recover something the watch forgot.

GUIDE 65W

Each of these four makes the next one readable, so ask what would be uninterpretable without it. A new sound is only new against a baseline. A missing tonal line means nothing until you know where the layer is. Bearing rates take minutes, so they cannot begin at the entrance. And one of these is agreed early for a different reason: afterwards, everybody is busy.

ASK 3W

Order the passage

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

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TAKES AS READ

a measurement can be meaningless without an earlier one to read it against

VERDICT 89W

A passage plan is a set of conditions that have to be true at the moment the boat commits. Each card is a prerequisite for reading the next. Without the own-noise baseline a new sound is just a sound. Without the layer depth a missing narrowband line looks like an absent contact rather than a bent path. Bearing rates take time, which is why they cannot be what you start collecting at the entrance. And the lost-contact agreement is made early for a different reason: afterwards, everybody is busy.

TAKEAWAY 15W

A passage plan is ordered by what each measurement needs in order to mean anything.

4.2 The margin that survives being wrong

NAVIGATION & POSITION · CONTROL ROOM · A PERSON · BALLPARK

SCENE 43W

The narrows fill the next chart panel. They are four nautical miles across, the plotted-position uncertainty is a circle about two miles wide, and the bank lies on the left side. The navigator asks for the surviving clearance before committing the centerline route.

GUIDE 65W

Four numbers, and one is own speed, which does not enter a clearance at all. Another is the number of halves, a step rather than a measurement. The circle around a plotted position is the honest width of where the boat might be. So the whole circle has to fit inside safe water. Four miles sounds generous until the uncertainty is on the same scale.

ASK 15W

How much clear water is left if the boat runs the middle of the channel?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a plotted position is the centre of a probability, not a fact

VERDICT 77W

The circle around a plotted position is the honest width of where the boat might actually be. Route planning therefore uses the circle, not just the pencil line at its centre. The whole uncertainty has to remain inside safe water. Half the channel is two miles. Half the position uncertainty is one mile. The surviving margin is the difference: one mile. Four miles of channel sounds generous until the position uncertainty is put on the same scale.

TAKEAWAY 12W

A route is chosen against the uncertainty, not against the plotted line.

4.3 The one shaft speed that rings

PROPULSION & ENGINEERING · PROPULSION MACHINERY · A ROOM · CHOICE

SCENE 30W

At 92 revolutions the after compartment rings loudly enough to hear through a bulkhead. At 85 and at 100 it does not. Nothing else about the plant changes between them.

GUIDE 74W

All four options are real machinery faults and three of them get steadily worse with speed. Ask of each whether it would come and go at one speed and nowhere else. Imbalance grows, bearings degrade, cavitation starts and stays. Only something that depends on matching a frequency can be loud at ninety-two revolutions and quiet at eighty-five and a hundred. That distinction decides whether the fix is a repair or an avoided speed band.

ASK 13W

The hull rings at one shaft speed and nowhere else. Why that one?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

every structure has frequencies at which it prefers to vibrate · sound as a wave:
frequency, wavelength, speed — taken as read

VERDICT 89W

A structure has natural vibration modes. If a repeating force lands close to one mode, each cycle adds energy in step with the motion and the vibration grows. Blade rate is shaft revolutions per second multiplied by the number of blades. One shaft speed can therefore put that forcing near a hull or mount resonance while nearby speeds do not. That is why the practical fix can be an avoided speed band or a

changed mount stiffness. The water-wave relation $v = f\lambda$ is not the governing equation here.

TAKEAWAY 21W

A structure has natural vibration frequencies, and forcing near one of them can make a small input produce a large response.

4.4 What a walkdown is for

DAMAGE CONTROL · FORWARD EQUIPMENT & HANDLING · A ROOM · CALLBACK · SEQUENCE

SCENE 42W

Chief Ferro walks you into the forward space and asks you to learn it. Equipment down both sides, deck plates over a recess, a hose reel on the bulkhead, two ways out. He is not telling you what to take in first.

GUIDE 67W

All four of these can be done in the same minute, so this is not a timetable. Ask instead what each fact lets you interpret. A locker of shoring means nothing until you know what this space can do to you. A hatch you have not tried is an assumption. At ninety metres, one of these decides whether a casualty is this compartment's or the next one's.

ASK 12W

Learn a compartment in the order the facts depend on each other

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a submarine is divided into compartments that can be sealed off

VERDICT 80W

A submarine survives a casualty by being divided. The pressure boundary comes first: at 90 metres it separates a flooded compartment from a flooded boat. The ways out come next, and they have to be proved rather than assumed. Then find what could start a casualty here: sea connections, cables or stores. Last, judge whether the equipment stored here can fight those threats. A locker is only the right locker once you know what this space can do to you.

TAKEAWAY 15W

A compartment is a pressure boundary first; everything you learn about it hangs off that.

Quiet costs capability, and the trade has to be made deliberately rather than by default.

Forward Flooding

PLAN CARD 97W

Sixth day, morning watch, and water is over the bilge in the forward space with the drain pump running. Ruth Hallam, on the pumps, has the level and reports it steady. Ferro does not believe steady means anything and is looking for the hole. This watch you find where the water is coming in, work out whether pumping can win, and decide what shutting that supply costs elsewhere — the two valves that stop it sit on a header that feeds other things. Pumping buys minutes. They are worth nothing unless somebody spends them finding the source.

BRIEFING 41W

The forward space is taking water. The drain pump is running and the level is still rising. Somewhere under those deck plates is a hole, and the boat is at 90 metres, which is what makes it a hole worth respecting.

WHAT YOU DECIDE 12W

Find the water, decide whether pumping can win, and stop the source.

5.1 Under the deck plates

DAMAGE CONTROL · FORWARD EQUIPMENT & HANDLING · A ROOM · BALANCE

SCENE 39W

The forward bilge alarm has sounded twice, and water is visible through the deck-plate slots. Hallam marks the level one minute apart while the drain pump runs. Fresh-water level and sonar-cooling flow stay steady. Command wants the inflow now.

GUIDE 61W

Reading a stream costs nothing, so read them all. Then count only the ones in the units command asked for: water entering the compartment, before the pump takes any out. The level is rising while the drain pump runs, so the pump belongs in the sum. Fresh-water level and sonar cooling are steady, which is information about what to leave out.

ASK 15W

How fast is water entering the compartment before the drain pump removes any of it?

BEHIND THE BUTTON 129W

Why the rise is not the inflow. What accumulates is what came in minus what has been removed. The pump runs throughout the minute Hallam is timing, so the observed rise already has the pump's contribution subtracted. Adding it back gives the number command asked for.

What counting a stream claims. Reading is neutral; counting says this quantity belongs in the same ledger. A steady fresh-water level and a steady cooling flow are not sources of the flooding, and including them would answer a different question.

Why the inflow rather than the level. The level says how bad it is now. The inflow says how long there is and whether the pumps can win, which is what the decision to blow ballast or shut the compartment is made on.

TAKES AS READ

a measured level rise is accumulation, not total inflow · a running pump removes water during the same measurement · pressure, depth and hydrostatics — taken as read

VERDICT 83W

The bilge rise shows only what stayed in the boat during the minute. It does not include water the drain pump removed at the same time. The correct balance is inflow = accumulation + removal. Here that is $88 + 55 = 143$ gallons per minute. The unchanged fresh-water tank and normal sonar-cooling flow contribute zero to the balance. A pump can make a serious leak look smaller on the deck plates, which is exactly why its discharge belongs back in the total.

TAKEAWAY 20W

Accumulation equals inflow minus removal, so a removal already running has to be added back to recover the true inflow.

5.2 What 90 metres puts behind the valve

DAMAGE CONTROL · FORWARD EQUIPMENT & HANDLING · A PERSON · BALLPARK

SCENE 40W

The suspected hull connection is 90 metres below the surface. Ferro writes seawater density as 1025 kilograms per cubic metre and g as 9.81 metres per second squared. He wants the gauge pressure on the sea side of the fitting.

GUIDE 78W

Five numbers, and they are not all about the same thing. Two describe the bilge: how fast the level rose, and what a centimetre of it holds. One is what the drain pump can shift at this head. One is the depth, and one is where the level started. Ask of each whether it describes the water coming in or the water going out. A rate the pump cannot beat is a different situation from one it can.

ASK 12W

About how much gauge pressure is the sea applying at 90 metres?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

gauge pressure counts the pressure added by the water column above the surface value · buoyancy and Archimedes — taken as read

VERDICT 78W

Hydrostatic pressure is $p = \rho gh$. Use seawater density 1025 kg/m³, $g = 9.81$ m/s², and $h = 90$ m. The gauge pressure is about 905,000 pascals, or 0.91 megapascals. That is roughly nine atmospheres above surface pressure. Add the atmosphere already present, and the absolute pressure outside the hull is about ten atmospheres. This is why a small opening at depth can drive a large inflow and why the hull isolation matters more than adding another pump.

TAKEAWAY 9W

Hydrostatic gauge pressure rises in direct proportion to depth.

5.3 What else that valve feeds

PROPULSION & ENGINEERING · PROPULSION MACHINERY · A ROOM · PROTOCOL

SCENE 35W

Two valves will stop the water: the outboard hull valve and the inboard supply valve. The header they sit on also feeds the sonar-array cooling. Haruki wants each action matched to what it actually achieves.

GUIDE 65W

Four actions, and every one of them takes something away from somebody. The header that feeds the break also cools the sonar array. Ask of each action which side of the hull it acts on, and what stops as a result. Only one cut separates the boat from the sea. Two of the others change how fast the compartment fills without changing whether it fills.

ASK 8W

Match each action to what it actually achieves.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

one header can serve more than one system · sound as a wave: frequency, wavelength, speed — taken as read · interference, beats and phase — taken as read

VERDICT 81W

Isolation is never free. Every valve shut can take another system away from somebody else. Here, stopping the flood and removing sonar-array cooling can happen through the same header. Only the hull valve separates the boat from the sea, so it is the cut that ends the source. The inboard valve stops flow to the break but leaves the sea connected to the header. A second pump buys time. Warning sonar changes nothing physical, but it lets the affected watch prepare.

TAKEAWAY 12W

Every isolation stops something else, and the order decides what that is.

5.4 The trip back without a waypoint

NAVIGATION & POSITION · CONTROL ROOM · A ROOM · CALLBACK · ROUTE

SCENE 39W

The navigator sends you from control to machinery with a sealed note. On the return trip one watertight door will be shut and the marker lights will be off. You have walked the route once, with labels still visible.

GUIDE 66W

You walk the route once with the labels showing. Then the labels go and you run it back from memory. One watertight door will be shut and the marker lights will be off. When that happens, find the landmark in the compartment you are actually in and work out where the detour has put you. You are graded on the compartments you reach, not the time.

ASK 22W

Run the route from memory. After the blocked door, use the next landmark to work out where the detour has put you.

BEHIND THE BUTTON 136W

Why places beat counting. A remembered count of turns survives only while nothing interrupts it. Miss one and every number after it is wrong, with nothing to tell you so. A place is different: two helm wheels means Control whether you arrived as planned or by a detour.

What the landmarks are for. Every compartment holds something that is nowhere else on the boat — the wheels, the radio racks, the long mess table, a particular pump. That is how a crew navigates in smoke or darkness, and it is what the second run is testing.

Why the door is shut without warning. Routes are learned in good conditions and used in bad ones. A route that only works when every door is open is not a route, it is a memory of one particular walk.

TAKES AS READ

a submarine is divided into named compartments · a route can be remembered as a sequence of places

VERDICT 73W

A count only works while nothing interrupts it. A turn list fails after one extra turn. A route made of places is different because each compartment becomes a new reference point. If you are diverted, identify the space you enter. Then rejoin the sequence from there. Submarines use the same logic when dead reckoning. The useful question is not how many steps have passed. It is what trusted reference remains after the interruption.

TAKEAWAY 18W

A route remembered as places can be recovered after an interruption; a count or turn list usually cannot.

END OF THE WATCH 11W

Flooding is stopped at the hull fitting; everything else is delay.

Electrical Fire

BEFORE THE WATCH — HUNT

Six breathing sets, and the locker list says four

Before the next drill every emergency breathing set has to be accounted for. Four are in the lockers where the list says. Two are not, because somebody moved them during the last maintenance period and told nobody. Find them, because a set nobody can locate is a set that does not exist when the compartment fills with smoke.

PLAN CARD 106W

Same day, afternoon watch, eight hours after the flooding and nobody has slept. Smoke from a switchboard in the electrical space. The indicator shows the bus open and a meter his own electrician's mate put on the cable inside reads 440 volts. The chief electrician, Bede Okonkwo, will not send anybody in on the strength of a lamp. This watch you take the fire apart into the things keeping it alive, decide what that meter is telling you, and work out how to fight a space nobody can see into. The order of your actions matters more here than the speed of any one of them.

BRIEFING 30W

Smoke in the electrical distribution space. The switchboard indicator says the affected bus is open, and the fire is behind a panel where nobody can see the seat of it.

WHAT YOU DECIDE 17W

Put out a fire whose ignition source is electrical, in the order that makes it stay out.

6.1 Three legs, three different actions

FIRE & ELECTRICAL · ELECTRICAL DISTRIBUTION · A ROOM · PROTOCOL

SCENE 33W

Chief Okonkwo puts the three legs of the fire on the board. The fault that keeps lighting it, the insulation that burns, the air in the compartment. He asks what removes each one.

GUIDE 65W

Three items on the left are legs of the fire. One is evidence about the space next door. Pair each by asking what it supplies: energy, fuel, or oxygen. Removing a leg is not the same as monitoring a symptom. An isolated cable can still be hot enough to relight. So proving a circuit dead and cooling the insulation are two actions with two tools.

ASK 12W

Match each leg of the fire to the action that removes it.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a fire needs fuel, oxygen and a source of energy

VERDICT 89W

An electrical fault can keep adding heat to burning insulation. The first electrical action is therefore to isolate the source and prove the conductor dead with a meter. That removes continuing electrical heating and the shock hazard. It does not guarantee that the fuel has stopped burning. The hot insulation must still be cooled below its re-ignition temperature. Ventilation is then controlled to limit oxygen supply and smoke movement according to the casualty plan. A hot bulkhead is evidence to monitor, not a fourth leg of the fire triangle.

TAKEAWAY 22W

For an electrical fire, de-energizing removes continuing electrical heating and shock hazard; the burning fuel still has to be cooled and controlled.

6.2 What a lamp is worth

FIRE & ELECTRICAL · ELECTRICAL DISTRIBUTION · A PERSON · SEQUENCE

SCENE 41W

Smoke is pushing through the compartment door. The switchboard indicator says the bus is open, but a meter on the cable reads 440 volts. The hose team is masked up at the threshold, waiting for your call before anyone steps inside.

GUIDE 70W

The switchboard lamp and the meter disagree, and the sequence has to survive that. Ask of each card what it proves rather than what it believes. A lamp reports the switchboard's opinion of the circuit. A meter reports the conductor. Water on a live cable carries current back to the hose team. They are masked up at the threshold, and this order is what stands between them and the fault.

ASK 6W

Prove it before you fight it

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a bus can be fed from more than one source

VERDICT 85W

The switchboard lamp reports what the switchboard believes about the circuit. A meter on the conductor reports the condition of that conductor. Those claims can disagree because contacts fail, lamps fail, or another source back-feeds the cable. Water on an energized conductor can carry current into the hose team. So the sequence matters: isolate the source, test the conductor, confirm zero voltage, then fight the fire. The meter is not extra paperwork. It is the check that separates an indicated isolation from a proved one.

TAKEAWAY 10W

An indicator reports a belief; a meter reports a condition.

6.3 Smoke is not only a breathing problem

ATMOSPHERE & LIFE SUPPORT · AUXILIARY MACHINERY & BILGE · A ROOM · CHOICE

SCENE 43W

Visibility is under half a metre. The far bulkhead is eight metres away. Temperature at head height reads 60 degrees against 28 at deck level. The team has air masks, and the fire is behind a panel where nobody can see it directly.

GUIDE 65W

All four options are true of a smoke-filled compartment in some way. They differ in what the smoke has taken away. Masks solve breathing and nothing else. Thirty-two degrees between deck and head height says the heat is layered. So conditions above the layer can worsen while the floor stays cool. The fire is behind a panel, so nobody will see it whatever they do.

ASK 11W

What does the smoke change about how the casualty is fought?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a fire inside a structure does not show you where it is · sound as a wave: frequency, wavelength, speed — taken as read

VERDICT 86W

Smoke changes more than breathing. It removes visibility, so the team can lose the route, the boundary and each other within a few metres. A large temperature difference between deck level and head height also shows strong thermal layering. Heat and smoke can therefore worsen above the layer even while the floor remains cooler. Air masks solve the breathing problem only. They do not restore sight or

remove heat. Entry now needs a controlled route, communications and a way to judge conditions without seeing the fire.

TAKEAWAY 13W

Smoke removes the instrument you were relying on, which is your own eyes.

6.4 Every fitting is an argument

PROPULSION & ENGINEERING · PROPULSION MACHINERY · A ROOM · CALLBACK · PROTOCOL

SCENE 39W

Chief Haruki stops beside four fittings on the same bulkhead and asks what each one is for. None carries a purpose label, only a number. He steps back from the valves and waits for you to make the distinctions.

GUIDE 67W

None of these four carries a purpose label, only a number. So work from shape, height and what lies beyond. Ask of each what is on the far side of it, and what opening it would admit or supply. The consequence is the clue. At ninety metres one of these has the sea behind it at ten atmospheres, and the wrong wheel is not a small mistake.

ASK 10W

Match each fitting to what it is there to do.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

pressure rises about one atmosphere for every ten metres of depth · pressure, depth and hydrostatics — taken as read

VERDICT 80W

A fitting makes sense only when you connect its shape to what lies beyond it. A wheel valve through the pressure hull may have the sea behind it at high pressure. A quick-connect manifold at head height can supply breathing air. A strapped locker can hold shoring and plugs. A hose reel can supply firefighting water. The point is not to memorize four silhouettes. It is to infer function from connection, location and the consequence of operating the wrong thing.

TAKEAWAY 20W

What a fitting is for is decided by what is on the other side of it, and at what pressure.

END OF THE WATCH 15W

An electrical fire is put out by de-energizing it; the agent only finishes the job.

Atmosphere Degradation

PLAN CARD 108W

Eighth day, middle watch. The scrubber plant is running and its outlet reads clean. The corpsman, Elena Navarro, has been round berthing with a handheld meter and it reads far higher than the sensor on that bulkhead, where nine people are asleep. The atmosphere control officer, Ana Lindqvist, has to decide whether to wake them. This watch you decide whether it is the air or the instrument that is wrong, work out how long that space is safe, and fix the circulation. This is the one system where the crew notice before the instruments do — headaches, short tempers, people asleep on watch — and they already have.

BRIEFING 34W

Two of the crew in berthing have headaches. The atmosphere plant says everything is normal. Both of those statements can be true at once, and working out how is the whole of this watch.

WHAT YOU DECIDE 14W

Tell a failing sensor from failing air, and find where the air stopped moving.

7.1 The compartment the scrubber cannot reach

ATMOSPHERE & LIFE SUPPORT · AUXILIARY MACHINERY & BILGE · A ROOM · CHOICE

SCENE 32W

The scrubber plant is running and its outlet reads clean. A handheld meter in berthing reads far higher than the sensor installed there, while control and machinery both read normal on handhelds.

GUIDE 59W

Four readings have to be explained at once. A clean plant outlet. A high handheld in berthing. A low installed sensor there. Normal handhelds everywhere else. Ask of each option how many of the four it covers. One fault does not have to explain everything when the evidence supports two. A scrubber can only treat air that reaches it.

ASK 6W

Which explanation accounts for every reading?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a plant can only treat air that reaches it

VERDICT 89W

The scrubber can only treat air that reaches it. A shut damper can therefore leave one compartment outside the circulation loop while the plant outlet still looks normal. The handheld reading in berthing is supported by symptoms there, while handhelds elsewhere are normal, so the problem is local rather than boat-wide. The installed berthing sensor also disagrees with the handheld and the symptoms, which points to a second fault in that sensor. One failure does not have to explain every bad reading when the evidence supports two independent faults.

TAKEAWAY 9W

Ventilation decides which compartments the plant is actually treating.

7.2 Nine people and a shut damper

ATMOSPHERE & LIFE SUPPORT · AUXILIARY MACHINERY & BILGE · A PERSON · BALLPARK

SCENE 44W

Navarro has 9 people in a 36-cubic-metre berthing space at 293 kelvin and about 1 atmosphere. With the damper shut, each person adds about 0.9 mole of carbon dioxide per hour. She wants the time for carbon dioxide to rise by one percentage point.

GUIDE 76W

Seven numbers, and one of them is the oxygen fraction, which this question is not about. Two are the crew: how many people, and what each produces per hour. Ask of each whether it belongs to the gas law or to the rate afterwards. The two halves are separate: first how many moles a one-point rise takes, then how long the crew needs to make them. A deadline in hours is what command can plan around.

ASK 14W

Using the ideal gas law, roughly how long does a one-percentage-point carbon-dioxide rise take?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

gas pressure, volume, amount and absolute temperature are related by $PV = nRT$ · the compartment temperature and total pressure stay roughly constant over the estimate

VERDICT 72W

A one-percentage-point rise means an added carbon-dioxide partial pressure of about 0.01 atmosphere, or 1013 pascals. Using $PV = nRT$, that is $n = PV/RT \approx 1013 \times 36 / (8.314 \times 293)$, about 15 moles. 9 people producing 0.9 mole per hour each add about 8.1 moles per hour. 15 divided by 8.1 is about 1.9 hours. The estimate is approximate, but it gives a planning deadline instead of the word “stuffy.”

TAKEAWAY 23W

The ideal gas law turns a concentration limit into a number of moles, and a production rate turns that amount into a deadline.

7.3 What one fan takes with it

PROPULSION & ENGINEERING · PROPULSION MACHINERY · A ROOM · CHOICE

SCENE 32W

The ventilation drawing shows one fan feeding the scrubber and the after spaces through a common trunk. The watch bill treats them as two independent systems, and the atmosphere plan assumes both.

GUIDE 71W

All four options treat one fan as a real constraint, and they differ in what they do about it. Ask of each whether it changes the number of ways the service can be lost. Running the scrubber harder does not add a path. Sealing the after spaces removes one need instead of adding redundancy. Two boxes on a watch bill are not independence; a different fan, power path or route is.

ASK 17W

One fan feeds both the scrubber and the after spaces. What does that mean for the plan?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a system is only as independent as the things it shares

VERDICT 80W

One fan can support two functions and still be one failure point. If the scrubber and after-space ventilation share the same fan and trunk, losing that fan removes both services together. Treating them as independent in the plan therefore overstates redundancy. The fix is not merely to write two boxes on the watch bill.

Independence requires a different fan, power path or airflow route. Until then, the atmosphere plan has one common-mode failure that can affect several compartments at once.

TAKEAWAY 14W

Two systems sharing one component are one system for the purpose of a failure.

7.4 Four traces, one of them ours

SONAR & ACOUSTICS · SONAR ROOM · A ROOM · CALLBACK · PROTOCOL

SCENE 42W

Four traces sit on the broadband display while the watch calls them out. The boat slowed four minutes ago, and one trace changed with it. The others kept their own patterns. Rask wants each source identified before the next sweep rolls across.

GUIDE 59W

Each trace carries more than one clue. How steady is the bearing? How wide is the spectrum? How long does it last? And did it follow the boat's own change of speed? Pair on the combination, not on which trace is loudest. The boat slowed four minutes ago, and that is a test one of these has already taken.

ASK 8W

Match each trace to what is making it.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

passive sonar listens and does not transmit · sound as a wave: frequency, wavelength, speed — taken as read

VERDICT 82W

Each trace has more than one clue. A sound that follows own-ship speed changes is probably generated aboard. A narrow line that drifts slowly in bearing fits steady machinery on another vessel. Short chirps that vanish quickly fit a biologic source better than a hull. A nearly steady bearing that grows louder deserves attention because its geometry is changing differently from the others. Classification comes from combining time, bearing, spectrum and response to own-ship changes, not from picking whichever trace looks strongest.

TAKEAWAY 12W

A source that changes when the boat changes belongs to the boat.

END OF THE WATCH 17W

A plant reading is a report about the plant, not about the compartment you are standing in.

Compound Casualty

PLAN CARD 106W

Ninth day, forenoon watch, and four things at once: water forward, a bus lost aft, the plot degraded by that power loss, and Adeyemi with a broken wrist in the passage. Vance wants to hold the ordered speed. Whitfield says a boat with four casualties making eight knots at a bank nine miles away is about to have five, and this time he is the one who is right. This watch you decide what you take, what you hand to whom, and whether the boat slows down. Everything you hold personally is something nobody else is working, and Whitfield will not invent the organisation for you.

BRIEFING 37W

Flooding forward, a partial power loss, a degraded plot and one injured crewman, all inside four minutes. There are not enough people to do everything, so the first decision is which things do not get done yet.

WHAT YOU DECIDE 12W

Prioritise and delegate when there are more casualties than there are people.

8.1 Four casualties, one crew

DAMAGE CONTROL · FORWARD EQUIPMENT & HANDLING · A ROOM · TRIAGE

SCENE 44W

Four reports hit control almost together. Water is forward and a bus is lost aft. The plot has degraded, and a crewman is down with a broken wrist. The alarm is still sounding. Every team leader is looking at you for the first move.

GUIDE 69W

All four of these are real and all four need owners. What separates them is how fast each changes while you deal with the others. Ask of each whether it grows by itself, and whether waiting makes it irreversible. A stable fracture hurts and does not spread. The boat is still closing on shallow water, so the plot has a clock too. It is not the fastest clock here.

ASK 5W

What gets the first team?

BEHIND THE BUTTON 194W

Why first is a different question from most important. When several calls compete for the same hour, what a choice is worth is not its own importance but the difference between doing it now and doing it later. A serious problem that will be no worse in an hour costs nothing to defer. A smaller one that closes a door costs everything behind that door.

What to look for in the options. Two things separate them: which is still changing, and which is a precondition for the others. A situation that is deteriorating has a cost per hour attached to it, and a step that unblocks the rest multiplies the value of the hours after it. Everything else is a preference about where to start.

Why only one answer is marked. In the situation these options describe all of them eventually happen; what is being tested is the head of the queue, because that is where the reasoning is visible. The verdict names what each of the others was waiting on, which is worth reading even when the choice was right — the ordering behind the first place is the rest of the answer.

TAKES AS READ

some problems get worse on their own and some do not

VERDICT 84W

Triage is about what changes fastest and what becomes irreversible, not what sounds worst. A growing flood can remove buoyancy and spread into new spaces. An electrical casualty can keep feeding heat or disable other systems. A degraded plot matters more as the boat approaches constrained water. A stable wrist fracture hurts and needs treatment, but it does not outrun the other three over the next minute. The first team goes where delay changes the future most; the rest still need owners and follow-up.

TAKEAWAY 15W

You order casualties by how fast they get worse, not by how loud they are.

8.2 Who does the things you are not doing

FIRE & ELECTRICAL · ELECTRICAL DISTRIBUTION · A PERSON · DELEGATE

SCENE 40W

The flooding is now falling because the pump removes 60 litres a minute against 40 coming in. A fire boundary aft is heating, the atmosphere is stable, and a crewman has a fractured wrist. The executive officer needs assignments now.

GUIDE 59W

Three problems and you keep one. For each of the others, name an owner, the first real action they will take, and the reading or event that brings it back to you. The flooding is now losing — sixty litres a minute out against forty in — so it is contained. The boundary aft is heating, which is not.

ASK 14W

Which problem do you keep, and what complete assignment does each remaining problem get?

BEHIND THE BUTTON 112W

How to rank the three. Trend, containment, consequence. Flooding that is losing is trending the right way. A heating fire boundary is trending the wrong way and has the worst consequence if it is ignored. A fractured wrist hurts and is stable.

Why a first action is required. An owner told to watch something comes back for instructions at the worst possible moment. Naming what they will do first is what lets them hold it without you.

Why a return condition is the other half. It is the number or event at which the problem stops being theirs, agreed now so nobody has to judge in the moment whether to interrupt you.

TAKES AS READ

a problem's trend matters as much as its present size · a complete delegation names an owner, a first action and a return condition · sound as a wave: frequency, wavelength, speed — taken as read

VERDICT 88W

The fire boundary is the one condition that gets worse fast enough to cross a damage threshold while you are deciding. At 54 °C and rising 6 °C per minute, an 80 °C insulation limit is only a few minutes away. The flooding is already losing 20 litres a minute net, so it can be watched. The atmosphere and wrist are stable. A useful handoff gives each problem an owner, a first action, and a number that brings it back to you. "Watch it" supplies none of those.

TAKEAWAY 17W

Delegation is complete only when an owner knows what to do first and when to report back.

8.3 Slowing down as a decision

NAVIGATION & POSITION · CONTROL ROOM · A ROOM · BALLPARK

SCENE 45W

The degraded plot is still being rebuilt while the boat makes eight knots toward a bank nine miles ahead. The officer of the deck can slow to three knots now. Whitfield wants to know how much thinking time that buys before anything harder to reverse.

GUIDE 64W

Five numbers, and two of them are not distances or speeds: the minutes since the plot degraded, and the difference between the two speeds. A difference in knots is not a difference in time, because time is distance divided by speed. Work out each speed's time and compare them. Slowing costs progress and buys everything else, and the passage will still be there afterwards.

ASK 10W

How much time does coming down to three knots buy?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

distance divided by speed is time

VERDICT 80W

Time is the only resource in a casualty that can be manufactured, and speed is the price list. Nine miles at eight knots is a little over an hour; the same nine miles at three knots is three. Slowing costs progress and buys everything else — the plot can be rebuilt, the flooding can be fought, the passage will still be there. Work out what eight knots is actually spending, in minutes, before deciding whether the boat can afford it.

TAKEAWAY 11W

Slowing the boat converts a distance problem into a time budget.

8.4 The trace that will not resolve

SONAR & ACOUSTICS · SONAR ROOM · A ROOM · CALLBACK · CHOICE

SCENE 33W

The fourth trace is faint, has held nearly the same bearing for 11 minutes, and does not appear on narrowband at all. The boat has not changed speed or depth since it appeared.

GUIDE 63W

Two facts have to be explained together: the trace is faint, and it has no narrowband line. Ask of each option whether it accounts for both. A layer can bend a path and make a source weak. It does not strip machinery tones away selectively. Eleven minutes is a long life for a processor artefact, and nothing aboard changed when the trace appeared.

ASK 10W

Which explanation fits every reading, not just the loudest one?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

sound bends when it crosses water of a different temperature

VERDICT 76W

Broadband measures sound energy across a band. Narrowband looks for stable tonal lines. A sound-speed layer can bend a path and make an external source weak, but it does not selectively strip away machinery tones. Faintness and missing tonals are therefore separate clues. The layer can explain the weak level; the absent narrowband says no tonal line is above threshold. 11 minutes of persistence argues against a transient artefact, and nothing aboard changed with the trace.

TAKEAWAY 16W

Refraction can weaken a received path, but it does not selectively erase the source's spectral content.

END OF THE WATCH 11W

Command is deciding what you do, and who does the rest.

Sonar Blinded by the Boat

BEFORE THE WATCH — CANVASS

Who actually heard the pipe

The order was piped through the boat twenty minutes ago and the coxswain does not believe it reached everybody. Ask along the compartments until enough answers have come back to say whether the pipe failed or whether the people in the after compartments were simply too busy to acknowledge it.

PLAN CARD 103W

Eleventh day. The broadband noise floor is up across every bearing rather than in 1 sector, and the broadband operator, Kes Dunne, has found a new narrowband line at 113 hertz that grows with turns. Rask says the boat is shouting into its own array; Haruki says his plant is running exactly as it was yesterday. This watch you decide whether the noise is outside the boat or inside it, narrow it to one machine, and say what kind of fault makes a line like that. While the array is degraded other boats can still hear this one, and it cannot hear them.

BRIEFING 37W

Sonar has lost the picture. There is a broadband rise across every bearing and a new narrowband line, and the ocean has not changed. The problem is aboard, which makes it the engineers' problem and sonar's evidence.

WHAT YOU DECIDE 15W

Find the noise the boat is making about itself, and prove which machine it is.

9.1 The rise on every bearing

SONAR & ACOUSTICS · SONAR ROOM · A ROOM · CHOICE

SCENE 38W

Broadband rises across every bearing at once. A new 212-hertz line also appears everywhere, and its frequency shifts slightly when the seawater-pump speed changes. The weather and sound-speed profile are steady. 1 pump was started 20 minutes ago.

GUIDE 65W

Two facts have to be explained together: a rise on every bearing at once, and a tonal line whose frequency follows the pump's speed. Ask of each option whether it has access to that control. An external source does not. Weather and sound speed do not track pump revolutions. A direction is what an outside contact occupies; the hull is what couples into every bearing.

ASK 6W

Which explanation fits the whole picture?

BEHIND THE BUTTON 195W

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TAKES AS READ

a contact out in the water lies in some direction · sound as a wave: frequency, wavelength, speed — taken as read

VERDICT 86W

An external contact occupies a direction; an own-ship source couples into the array from the hull and can contaminate many bearings at once. The decisive clue is not simply that the line is loud. Its frequency changes when the pump speed changes. Something outside the boat has no access to that control. Weather or a sound-speed change would not track pump revolutions either. The broad rise and control-linked

tonal line therefore point to machinery aboard the boat, which is both masking contacts and increasing radiated signature.

TAKEAWAY 13W

A signal that follows an own-ship control is evidence of an own-ship source.

9.2 Narrowing it to one mount

PROPULSION & ENGINEERING · PROPULSION MACHINERY · A ROOM · CONTROL

SCENE 42W

4 machines are running when a new 212-hertz line appears across the array. Only the seawater pump was started in the last half hour. Haruki will let you change one machine at a time, but sonar must report before each next change.

GUIDE 73W

The number you are watching is the strength of the 212-hertz line on the sonar array. Four machines are running, and only one of them has to be the source. Stop a machine and sonar reports again: a machine that matters takes the line down with it, and a machine that does not leaves it where it was. Restart what you stopped before the next change, then name the machine the line follows.

ASK 6W

Which machine causes the 212-hertz line?

BEHIND THE BUTTON 174W

Why one machine at a time. Two machines stopped together and a line that disappears tells you nothing about which one did it, and on a boat there is a further cost: every machine you leave stopped is a system somebody else needs. The rule is also a rule about the boat, not only about the measurement.

Why the timing is suggestive and not evidence. The seawater pump was started in the half hour before the line appeared, which makes it the obvious suspect and not a proven one. Plenty of things change in half an hour aboard a submarine, and a coincidence in time is exactly what this format exists to separate from a cause.

What the line is. A rotating machine radiates at frequencies set by its shaft rate and the number of blades or vanes, and a mount that has loosened couples that sound into the hull far more efficiently than it should. So the tone is a fingerprint: its frequency says what kind of machine, and switching says which one.

TAKES AS READ

changing several variables at once makes a causal result ambiguous · a suspected cause should reproduce the effect when restored · sound as a wave: frequency, wavelength, speed — taken as read

VERDICT 82W

The log tells you what changed near the time the line appeared, but timing alone is only a correlation. A causal test holds every other machine fixed, changes one control, and waits for sonar to report. The seawater pump changes the line by 13 dB while measurement noise is only 2 dB. The other machines move it by no more than the noise. Restoring the pump brings the line back. That reversal is what turns a plausible suspect into the measured cause.

TAKEAWAY 22W

A correlation becomes a causal test when one variable changes, the response follows, and the response returns when the variable is restored.

9.3 What the shifted line says about speed

SONAR & ACOUSTICS · SONAR ROOM · A PERSON · BALLPARK

SCENE 41W

After the 212-hertz own-ship line is removed, a separate contact on bearing 064° resolves at 149.7 hertz. Boats of that class produce the same machinery line at 150.0 hertz. The operator asks what the small frequency shift says about range rate.

GUIDE 66W

Four numbers, and one is a conversion into knots rather than a term. Note which frequency is the reference and which is the measurement. Their difference is what carries the information, and the sign matters as much as the size. Note whose line this is now. The own-ship tonal was removed first, because a machine aboard shares the receiver's motion and shows no Doppler at all.

ASK 10W

Estimate the range rate implied by the 0.3-hertz downward shift.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

sound travels at about 1500 metres a second in sea water · sound as a wave:
frequency, wavelength, speed — taken as read

VERDICT 80W

For speeds much smaller than sound speed, the fractional Doppler shift is approximately $\Delta f/f = v/c$. Here the observed line is 0.3 Hz below a 150.0 Hz rest line, so $\Delta f/f = -0.3/150 = -0.002$. Multiplying by 1500 m/s gives about -3 m/s, or roughly -5.8 knots. The negative sign means the range is opening. This calculation now belongs to an external contact; an own-ship machine would share the receiver's motion and would not show a six-knot relative Doppler shift.

TAKEAWAY 18W

A one-way Doppler shift measures line-of-sight relative speed; the sign tells whether the range is opening or closing.

9.4 Ours, or somebody else's

SONAR & ACOUSTICS · SONAR ROOM · A PERSON · BELT

SCENE 30W

The night's held contacts have to be logged before the watch changes. Some of what the array heard came from this boat. The rest came from the water outside it.

GUIDE 55W

Two bins. A sound made by this boat rides with it — the bearing does not change however the boat turns, and it comes and goes with what the engineers are doing. A sound from outside has a bearing of its own. Sort on where the sound is made, not on what it sounds like.

ASK 12W

Send each sound to the bin that says where it was made.

BEHIND THE BUTTON 87W

Why own noise is the hard case. It arrives on the same array as everything else and can be louder than a real contact. What separates it is that it does not move: turn the boat thirty degrees and an outside contact's bearing changes, and the pump's does not.

Why it matters for the log. Own noise logged as a contact puts a ghost on the plot that the next watch has to disprove. An outside contact logged as own noise is a boat nobody is tracking.

TAKES AS READ

a hydrophone hears the boat it is mounted on as well as the sea

VERDICT 165W

The array cannot tell where a sound was made; the geometry can. A sound generated on this boat arrives at the array from the same relative direction whatever the boat is doing, so the bearing follows the hull round a turn. A sound made in the water holds its own bearing while the boat turns under it, which is why a thirty-degree course change is the standard test and why it is worth the time it costs. The second clue is correlation: own noise starts and stops with something the crew did — a pump cut in, a shaft order, a hatch. A biologic or another vessel has no reason to be synchronised with the engine room. Getting the bin wrong costs in both directions. Own noise in the log is a ghost the next watch has to spend hours disproving; a real contact filed as own noise is a vessel nobody is tracking, on a night when the boat is trying to stay unheard.

TAKEAWAY 12W

A sound that turns with the boat was made by the boat.

9.5 The bearing that does not move

NAVIGATION & POSITION • CONTROL ROOM • A ROOM • CALLBACK • CHOICE

SCENE 42W

The contact has stayed within half a degree of the same bearing for six minutes. Its level rises on every sweep. The boat is still making eight knots, and the officer of the deck wants an answer before the channel narrows ahead.

GUIDE 69W

A bearing is a direction and not a range, so no option here can be settled by bearing alone. Ask what each one would require you to already know. Steady bearing with a rising level does not prove a collision. What it does is remove the comfortable reading, that the contact is sliding safely across the bow. The channel narrows ahead and the boat is still making eight knots.

ASK 11W

What does a steady bearing and a rising level tell you?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

passive sonar gives a direction and no range at all • sound as a wave: frequency, wavelength, speed — taken as read

VERDICT 82W

A bearing is the direction to a source, not its range. If the bearing stays nearly fixed while the sound grows stronger, the relative geometry is not opening sideways very quickly. That can happen when two tracks are converging. The important point is not that a steady bearing proves collision by itself. It is that steady bearing plus rising level removes one comforting explanation: the contact is not simply sliding safely across the bow. The watch needs range or motion information next.

TAKEAWAY 15W

A constant bearing with a rising level is a collision geometry, not a quiet one.

END OF THE WATCH 15W

Own-ship noise is diagnosed by correlating what sonar hears with what the plant is doing.

Uncontrolled Depth Change

PLAN CARD 106W

Twelfth day, middle watch. The boat is going deeper with the planes already at rise, the trim shows heavy forward, speed is unchanged, and the depth gauge is the calmest instrument in the control room. Ferro is on his way forward without being asked, which tells you what he thinks it is. This watch you decide what is actually causing it, work out how much weight is aboard, and choose which control to use — knowing two of the four are hard to undo at this depth. Treat the symptom and the boat keeps going down while everybody watches the right gauge for the wrong reason.

BRIEFING 37W

The boat is heavier than it was an hour ago and it is going down. The depth gauge, the trim state and the planes all have something to say and only one of them is the cause.

WHAT YOU DECIDE 16W

Tell a symptom from a cause when the boat is going somewhere you did not order.

10.1 Three gauges, one problem

NAVIGATION & POSITION · CONTROL ROOM · A ROOM · CHOICE

SCENE 43W

The depth needle is still moving deeper even though the planes are already at rise. Trim shows the boat heavy forward, speed has not changed, and the aft depth gauge disagrees with control by four metres. The next depth report is due now.

GUIDE 66W

Three gauges disagree and one cause has to cover all of them. The planes are already at rise. The trim is heavy forward. Four metres separate the two depth gauges. Ask of each option how many of those it explains. A gauge fault explains one. Neither depth reading is truth while they disagree, so the answer has to separate what is real from what is indication.

ASK 6W

Which cause accounts for every reading?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

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TAKES AS READ

a submarine holds depth by buoyancy and by the planes together

VERDICT 81W

A depth change can come from real vertical motion, a bad indication, or both. The planes already at rise say the control system is trying to oppose the descent. Heavy-forward trim is independent evidence that extra weight or lost buoyancy may be forward. The two depth gauges disagree, so neither should be treated as unquestioned truth. The readings together point to a real trim problem plus an indication problem. Command needs an independent depth check while the weight source is investigated.

TAKEAWAY 11W

The instrument that disagrees is a candidate, not automatically the culprit.

10.2 What 1.6 cubic metres of seawater is worth

DAMAGE CONTROL · FORWARD EQUIPMENT & HANDLING · A ROOM · BALLPARK

SCENE 42W

The forward bilge geometry gives about 1.6 cubic metres of seawater aboard. Ferro wants the force scale before command chooses how much ballast water to expel. Use seawater density 1025 kilograms per cubic metre and $g = 9.81$ metres per second squared.

GUIDE 74W

Four numbers, and one of them is what a tonne weighs, which is a conversion rather than a measurement of this flood. Ask of each whether it describes the water aboard or the units the answer is wanted in. A submerged boat has no simple tonnes-per-metre rule, so the depth change is not the measurement here. The volume of water actually aboard is, and command is choosing how much ballast to expel from it.

ASK 17W

What weight does 1.6 cubic metres of seawater add, and what equal buoyant-force scale would offset it?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

floodwater volume can be estimated from the bilge geometry · 1 tonne of mass weighs about 9.81 kilonewtons

VERDICT 82W

A fully submerged boat does not have a simple “tonnes per metre of depth” rule. Extra floodwater adds weight; depth change also depends on speed, planes and dynamics. Use the measured water volume instead. For 1.6 m³ of seawater, $\rho V g = 1025 \times 1.6 \times 9.81 \approx 16,100$ N. That is the weight of about 1.64 tonnes of seawater. Recovering an equal force through ballast action gives command the right response scale. A four-metre depth change alone does not measure added mass.

TAKEAWAY 21W

For seawater, $\rho V g$ gives both the weight of a known water volume and the buoyant force on an equal displaced volume.

10.3 What each control actually does

PROPULSION & ENGINEERING · PROPULSION MACHINERY · A ROOM · PROTOCOL

SCENE 45W

The depth is still increasing and 4 actions are available in the next 30 seconds. Planes, speed, ballast air and the trim pump can all change the boat. Two actions are hard to take back at this depth, and command wants their effects named first.

GUIDE 75W

Four actions in thirty seconds, and two of them are hard to take back at this depth. Pair each by asking what it changes: the lift the planes can make, or the weight the boat is carrying. Then ask what it costs. One removes weight in seconds and is heard for miles. One removes it quietly, and only if the pump can outrun the leak. One of the four moves the boat the wrong way.

ASK 8W

Match each action to what it actually achieves.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

the planes only work while water is flowing past them

VERDICT 89W

The controls do different jobs and have different costs. More speed increases flow over the planes, so the same rise angle produces more hydrodynamic lift. Blowing main ballast removes water quickly and powerfully, but it spends stored air and creates a strong acoustic signature. Pumping a trim tank to sea removes weight more slowly and can be quieter, if the pump outruns the leak. Slowing reduces flow over the planes and therefore reduces their immediate authority. Matching action to effect keeps an urgent response from becoming an irreversible guess.

TAKEAWAY 12W

Speed and angle buy time; blowing buys depth and gives away position.

10.4 Ordered depth, while the boat changes weight

PROPULSION & ENGINEERING · PROPULSION MACHINERY · A PERSON · HOLD

SCENE 37W

The boat has to hold sixty metres while the galley pumps out, the torpedo tubes are flooded for a drill, and the sea gets colder over the shelf. The trim pump is what answers all of it.

GUIDE 59W

Hold the boat at ordered depth, inside the band on the gauge. The band narrows as the passage goes on, because a boat under a layer has less room to wander than one in open water. The trim pump moves water between tanks and the sea, and every change in the boat's weight keeps pushing until you answer it.

ASK 12W

Hold sixty metres while the boat's weight and the water both change.

BEHIND THE BUTTON 152W

Why a boat drifts off depth at all. A submarine at ordered depth is balanced: the weight of the boat against the weight of the water it displaces. Pump a tonne of galley waste over the side and the boat is a tonne light, so it rises — and goes on rising.

Why the sea itself moves you. Colder, saltier water is denser, so the same boat displaces more weight in it and floats up. Crossing a front is a change in the water rather than in the boat, and it feels identical on the gauge.

Why the pump has to be set rather than tapped. Every one of these is a step in the rate. A boat a tonne light does not rise by two metres and stop; it keeps rising until the trim is put back, which is why the answer is a new pump setting held for a while.

TAKES AS READ

a submarine is balanced between its weight and the water it displaces

VERDICT 153W

A submarine at ordered depth is a balance between what it weighs and what it displaces, and everything in this passage disturbs one side of that. Pumping the galley over the side makes the boat lighter, so it rises and keeps rising. Flooding the tubes makes it heavier, so it sinks and keeps sinking. Crossing onto the shelf changes neither: the water gets denser, the same hull displaces more weight, and the boat floats up exactly as though somebody had pumped. All three are steps in the rate rather than steps in the depth, which is why the trim pump is set to a new rate and held rather than tapped and released. The band narrowing is the other half — under a layer, holding depth is what keeps the boat quiet and keeps the sonar picture usable, so the same skill has to be exercised more tightly exactly where it is hardest.

TAKEAWAY 15W

A boat that has changed weight is not a boat that has changed depth once.

10.5 Where the sound will not go

SONAR & ACOUSTICS · SONAR ROOM · A ROOM · CALLBACK · CHOICE

SCENE 39W

The sound-speed trace reads 1505 metres per second above 60 metres and 1485 below it. A contact held for 40 minutes while this boat stayed at 20 metres, then vanished in 1 sweep while every other trace remained unchanged.

GUIDE 63W

The contact vanished in one sweep while every other trace held. Ask of each option whether it explains that suddenness. Range and shutdown both take effect gradually. The baffles are a direction rather than an event. The sound-speed numbers are what do the work here. Rays bend toward slower water, and a region can be left that rays from this depth never reach.

ASK 16W

The contact was held on the surface layer and is now lost. Where has it gone?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

Snell's-law angles are measured from the normal to the layer · a lower sound speed bends a ray toward the normal

VERDICT 74W

At a layer, Snell's law gives $\sin\theta_1/v_1 = \sin\theta_2/v_2$. For a ray at 60° from the vertical, $\sin\theta_2 = (1485/1505)\sin 60^\circ$, so θ_2 is about 58.7° . The smaller angle means the ray bends more toward the vertical as it enters slower water. Across a real sound-speed profile, repeated bending can leave a region that rays from this depth do not reach. A contact entering that shadow can disappear abruptly instead of fading smoothly with range.

TAKEAWAY 15W

Sound bends toward slower water, which leaves regions a sonar cannot hear into at all.

END OF THE WATCH 17W

Depth is the outcome of weight, angle and speed, and each of those has its own cause.

The Cooling Path

PLAN CARD 101W

Thirteenth day, and this one was bought two watches ago. A main bearing is running hot: the oil is hot, the fresh water loop is hot, and the seawater leaving the heat exchanger is barely warmer than the seawater going in. The second cooling pump is still secured from the quiet lineup Rask asked for, and its breaker is open at the switchboard with nothing in the log. This watch you write the cooling path down in order and decide which link stopped handing the heat on. A bearing run hot for an hour is a bearing replaced in a shipyard.

BRIEFING 39W

Lube oil temperature is rising on the main bearing and nothing has changed on the throttle. Somewhere between the sea and that bearing there is a chain of things that carry heat away, and one of them is not.

WHAT YOU DECIDE 12W

Trace one cooling path end to end and find where it stopped.

11.1 From the sea to the bearing

PROPULSION & ENGINEERING · PROPULSION MACHINERY · A ROOM · CHAIN

SCENE 38W

The bearing is running warm at its usual load. Haruki puts the cooling diagram on the table before anyone touches a valve: bearing, lube oil, oil cooler, fresh-water loop, seawater cooler, and overboard discharge. One link is failing.

GUIDE 56W

Put the six stages in the order the heat actually travels, from the bearing out to the sea. Then read what each handover is doing and name the one that is limiting the chain. The bearing is hottest because it is where the heat is made; that would be true even if everything downstream were perfect.

ASK 13W

Where does the heat travel, and which required transfer is limiting the chain?

BEHIND THE BUTTON 137W

What each handover is. The bearing makes heat. Lube oil carries it away. The oil cooler hands it to the fresh-water loop, the seawater cooler hands that to seawater, and the discharge carries it overboard. Every one of those has to pass on as much as arrives, or heat backs up behind it.

How to find the limiting one. Compare what each stage is receiving against what it is passing on. The stage that has fallen behind — a fouled cooler, a throttled valve, a low flow — is the one setting the temperature everywhere upstream of it.

Why the diagram comes out before anybody touches a valve. Changing a flow in the middle of a chain moves the symptom without fixing the cause, and on a submarine an unnecessary valve movement is also a noise transient.

TAKES AS READ

heat must be transferred through every required link before it can leave the boat • a hot component is not automatically the failed link

VERDICT 84W

The bearing creates heat, but that does not make the bearing the failed link. Heat must move into the lube oil, across the oil cooler, through the fresh-water loop, across the seawater cooler, and finally overboard. If any required handoff is weak, everything upstream gets hotter. The current readings show normal bearing load and normal oil-side flow, but reduced seawater flow. That makes the final rejection path the governing link. Writing the chain first prevents the hottest component from being mistaken for the cause.

TAKEAWAY 18W

A transfer chain is limited by the weakest required handoff, not necessarily by the hottest or largest component.

11.2 One link is not handing the heat on

PROPULSION & ENGINEERING · PROPULSION MACHINERY · A PERSON · CHOICE

SCENE 44W

The lube oil and fresh-water loop are both hot. Seawater enters the cooler at 9 °C and leaves at 23 °C; its normal outlet is about 12 °C at this load. The seawater flow indicator reads 45%, and 1 of 2 pumps is secured.

GUIDE 63W

Follow the heat along the chain and ask where it stops being handed on. Two facts constrain it. The fresh-water loop is hot, so heat is reaching the cooler. The seawater leaves fourteen degrees warmer instead of three. Ask of each option what it would do to that outlet temperature. Poor transfer across the exchanger gives a small rise, not a large one.

ASK 7W

Which link in the chain has failed?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a heat exchanger needs flow on both sides to move heat

VERDICT 81W

Heat carried by a coolant depends on mass flow and temperature rise. If the heat load stays similar and flow falls, each kilogram carries more energy, so ΔT grows. Here seawater warms 14 °C instead of about 3 °C while flow is only 45% of normal. The hot fresh-water loop shows heat reaches the seawater cooler. The large seawater temperature rise shows too little water is carrying it away. A small outlet rise would instead suggest poor transfer across the exchanger.

TAKEAWAY 23W

For a similar heat load, lower coolant mass flow makes each kilogram of coolant warm more, so the outlet temperature rise gets larger.

11.3 The breaker nobody logged

FIRE & ELECTRICAL · ELECTRICAL DISTRIBUTION · A ROOM · CHOICE

SCENE 33W

The second seawater pump did not trip; its breaker is open at the switchboard. The quiet-running lineup from two watches ago is still in force and nobody wrote down what it turned off.

GUIDE 67W

All four options are defensible readings of an open breaker. They differ in where the failure was. Ask of each whether it addresses the trade itself or the record of the trade. Securing the pump was a reasonable decision at the time. What did not survive the watch change was why. A later watch sees an open breaker and cannot tell a valid lineup from a fault.

ASK 8W

What does this say about the quiet-running lineup?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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TAKES AS READ

a watch hands over to a watch that was not there

VERDICT 83W

The quiet-running lineup was a deliberate trade, but the reason for that trade did not survive the watch change. A log has to carry more than the fact that a machine was secured. It must also say what that machine supports and when it should return. Without that context, a later watch sees an open breaker and cannot tell a valid configuration from a fault. The cooling problem is therefore also a handover problem: the plant state outlived the decision that created it.

TAKEAWAY 10W

A configuration nobody recorded is a configuration nobody will restore.

11.4 How far the water has carried us

NAVIGATION & POSITION · CONTROL ROOM · A ROOM · CALLBACK · BALLPARK

SCENE 35W

The last trusted fix was three and a half hours ago. The forecast set for this area is about half a knot, and nobody has applied it to the plot because nobody has measured it.

GUIDE 64W

Five numbers, and three belong to the boat rather than the water: own speed, depth and course. Ask of each whether it moves the boat through the water or carries the water itself. Speed through the water cannot see a current. Half a knot for three and a half hours is nearly two miles. This is the plot the boat enters the narrows on.

ASK 14W

Estimate how far the boat could be from where the plot says it is.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

the log measures speed through the water, not over the ground

VERDICT 78W

Between fixes, the boat carries its position forward from the last trusted point using course, speed and time. That is dead reckoning. The water moves too, and a current can carry the whole boat sideways without appearing in speed through the water. Half a knot for three and a half hours is nearly two miles of set. The error grows every hour it goes unmeasured. This is the same plot the boat will use to enter the narrows.

TAKEAWAY 12W

Dead reckoning tells you where you would be if nothing pushed you.

END OF THE WATCH 14W

A cooling system is a chain, and a chain is diagnosed link by link.

Silent-Running Configuration

BEFORE THE WATCH — EVADE

The chief wants a decision you are not ready to make

The chief has been looking for you since the watch turned over, wanting a verbal answer on the port motor that you have not got the readings for yet. Stay out of his way until the readings are in. An answer given in a passageway is an answer nobody can produce afterwards, and this boat runs on what is written in the log.

PLAN CARD 101W

Fifteenth day, and the argument that has been running all patrol comes due. Four hours of quiet running, and Haruki intends to secure everything on Rask's list to get it — including the second seawater pump and half the carbon dioxide scrubbing. Lindqvist has done the arithmetic on the second one and is not happy. This watch you decide what may be shut down, what may only be slowed, and what the whole configuration costs the boat. Quiet is bought with capability, and the bill arrives late, in another department. None of that appears on the same gauge as the decision.

BRIEFING 32W

The boat has to be quiet for four hours. Everything aboard that makes noise also does something, and the ones that keep people breathing and bearings cool are not on the table.

WHAT YOU DECIDE 13W

Decide what may be secured, reduced or rescheduled while keeping the boat alive.

12.1 Four hours of quiet

PROPULSION & ENGINEERING · PROPULSION MACHINERY · A ROOM · STRESS

SCENE 38W

2 narrowband components sit at 147 and 150 hertz on the same bearing. In seawater the sound speed is about 1500 metres per second. The audio level swells and fades regularly while both spectral lines themselves remain steady.

GUIDE 61W

Move the two-hour schedule margin down to zero and watch which lineups go dark. A lineup that works only with margin in hand is not a lineup. Sound speed in seawater is about 1500 metres a second, and both lines stay steady while the level swells and fades — those are the facts the surviving plan has to be consistent with.

ASK 14W

Which lineup survives when the two-hour schedule margin falls all the way to zero?

BEHIND THE BUTTON 137W

What the swelling and fading is. Two steady tones close in frequency add and cancel as they drift in and out of phase, at a rate set by their difference — three hertz here, so a swell every third of a second. It sounds like a machine changing load and it is two machines that are not.

Why the margin is the thing to stress. A schedule with two hours spare hides a lot: a slow transit, a delayed evolution, a repair that runs long. Testing at zero is what separates plans that work from plans that have been lucky.

Why sound speed appears in a scheduling problem. It sets how long the boat is detectable at a given range as the geometry changes, so the acoustic constraint and the timetable are the same constraint written twice.

TAKES AS READ

a robust plan is tested at the pessimistic end of a stated uncertainty · quieting measures can trade acoustic benefit against equipment endurance

VERDICT 78W

A four-hour transit leaves two hours before the six-hour planning ceiling. Maximum Quiet looks best there because it cuts noise by 12 dB. But its equipment clock lasts only five hours, so it needs at least one hour of schedule margin. Balanced Quiet reaches the 7 dB requirement and lasts 8 hours, so it still works with no margin left. Conservative lasts, but is too loud. Moving the margin changes the winner because uncertainty belongs inside the decision.

TAKEAWAY 13W

A robust choice still works when the margin you thought you had disappears.

12.2 How long one scrubber lasts

ATMOSPHERE & LIFE SUPPORT · AUXILIARY MACHINERY & BILGE · A PERSON · BALLPARK

SCENE 33W

With one scrubber train secured, the remaining train removes less carbon dioxide than the crew produces. Lindqvist wants the number of hours before the boat has to make a noise to fix that.

GUIDE 62W

Four numbers. One is the length of the transit, which the answer is compared against rather than built from. Two are rates that point in opposite directions. Ask of each whether it adds carbon dioxide, removes it, or measures the room left. A net rate is the difference between two of these. A deadline in hours is what command can plan around.

ASK 10W

How long can the boat run on one scrubber train?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a net rate is what is produced minus what is removed

VERDICT 84W

A closed atmosphere with a production rate and a removal rate has a net rate. Compare that net rate with the remaining headroom to get a deadline. Here the crew produces 1.4 cubic metres per hour while 1 train removes 1.1. The net rise is 0.3 cubic metres per hour. 6 cubic metres of headroom therefore lasts about 20 hours. Command can plan around 20 hours. It cannot plan around “soon,” which is why a slow problem becomes useful once its rate is known.

TAKEAWAY 17W

A slow problem with a known rate is a deadline, and a deadline can be planned around.

12.3 What silence buys, term by term

SONAR & ACOUSTICS · SONAR ROOM · A ROOM · BALLPARK

SCENE 37W

After the quiet lineup, target source level is 145 dB and one-way transmission loss is 55 dB. Ambient noise is 85 dB, with 20 dB of array gain. The operator needs 10 dB signal-to-noise ratio for detection.

GUIDE 70W

Six numbers, and one of them is target strength, which belongs to an echo. This is a passive listen, so nothing here comes back. Ask of each term whether it helps the signal or hurts it, and how many times the sound crosses the water. Transmission loss appears once, not twice. The threshold is what the answer is measured against, so keep it separate from the budget that produces it.

ASK 11W

How many decibels of detection margin does the current lineup have?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

passive sonar receives a one-way signal from a source · array gain reduces the effective noise seen by a directional receiver · decibels and logarithmic scales — taken as read

VERDICT 91W

Passive sonar listens to a source rather than an echo, so transmission loss appears once and there is no target-strength term. The budget is $SNR = SL - TL - (NL - AG)$. Here that is $145 - 55 - (85 - 20) = 25$ dB. The operator needs 10 dB, so the signal excess is 15 dB. This is different from the active echo equation, which has two transmission-loss legs and a target-strength term. The distinction matters because the old equation badge was an active-sonar formula on a passive-sonar lesson.

TAKEAWAY 21W

Passive detection is a one-way budget: source level minus one transmission loss and effective noise, with array gain helping the receiver.

12.4 What the picture is for tonight

NAVIGATION & POSITION · CONTROL ROOM · A ROOM · SPOT

SCENE 43W

Contacts come onto the board through the watch. The officer of the watch holds one standing order about which of them matter, and rewrites it as the passage changes — after the fishing fleet, after the range closes, after the boat goes quiet.

GUIDE 57W

Hold the contacts the standing order currently wants and let the rest go. The order is changed during the watch and nobody announces it. What is scored is the contacts either side of a change, because they are the only ones that show whether the plot is being kept to tonight's order or to the last one.

ASK 11W

Hold the contacts tonight's order wants, and keep watching the order.

BEHIND THE BUTTON 98W

Why a standing order changes. In open water the plot is about anything that could close the boat. Through a fishing fleet it is about anything under way rather than drifting. Once the boat is rigged for quiet it is about anything that could hear us, which is a different list again.

Why the changeover is where a plot goes wrong. A team holding to the old order for ten more minutes drops the contacts the new order was written to keep, and a contact dropped from the plot leaves nothing behind to say it was ever there.

TAKES AS READ

a watch works to a stated order about which contacts are held

VERDICT 118W

Three orders run across the watch and each wants a different part of the board. Anything closing, then anything under way, then anything with active sonar. A contact can answer two at once, which is what makes the switch cost real. The panel scores the window either side of each change, because most of what is on the board is wanted by neither order and is correctly let go by somebody who has read nothing. The control room's own version is the ten minutes after the boat is rigged for quiet: the order has changed to anything that could hear us, and the plot being maintained is still the one that was tracking anything that could hit us.

TAKEAWAY 15W

The cost of a withdrawn instruction is paid by whoever is still working to it.

12.5 Under the deck plates

DAMAGE CONTROL · FORWARD EQUIPMENT & HANDLING · A ROOM · CALLBACK · BALANCE

SCENE 39W

The forward bilge alarm has sounded twice, and water is visible through the deck-plate slots. Hallam marks the level one minute apart while the drain pump runs. Fresh-water level and sonar-cooling flow stay steady. Command wants the inflow now.

GUIDE 61W

Reading a stream costs nothing, so read them all. Then count only the ones in the units command asked for: water entering the compartment, before the pump takes any out. The level is rising while the drain pump runs, so the pump belongs in the sum. Fresh-water level and sonar cooling are steady, which is information about what to leave out.

ASK 15W

How fast is water entering the compartment before the drain pump removes any of it?

BEHIND THE BUTTON 129W

Why the rise is not the inflow. What accumulates is what came in minus what has been removed. The pump runs throughout the minute Hallam is timing, so the observed rise already has the pump's contribution subtracted. Adding it back gives the number command asked for.

What counting a stream claims. Reading is neutral; counting says this quantity belongs in the same ledger. A steady fresh-water level and a steady cooling flow are not sources of the flooding, and including them would answer a different question.

Why the inflow rather than the level. The level says how bad it is now. The inflow says how long there is and whether the pumps can win, which is what the decision to blow ballast or shut the compartment is made on.

TAKES AS READ

a measured level rise is accumulation, not total inflow · a running pump removes water during the same measurement · pressure, depth and hydrostatics — taken as read

VERDICT 83W

The bilge rise shows only what stayed in the boat during the minute. It does not include water the drain pump removed at the same time. The correct balance is $\text{inflow} = \text{accumulation} + \text{removal}$. Here that is $88 + 55 = 143$ gallons per minute. The unchanged fresh-water tank and normal sonar-cooling flow contribute zero to the balance. A pump can make a serious leak look smaller on the deck plates, which is exactly why its discharge belongs back in the total.

TAKEAWAY 20W

Accumulation equals inflow minus removal, so a removal already running has to be added back to recover the true inflow.

END OF THE WATCH 13W

A silent lineup is a set of deadlines, not a set of switches.

Rig for Dive

PLAN CARD 106W

Nineteenth day. The boat has been open to the surface since Tuesday for work on the mast and dives at dawn. The switchboard shows two breakers open from that work, the log says one was restored, and every valve on the rig sheet has signatures against it — including one from a man who was ashore that morning. This watch you work out what each checklist line is really asking, close the boat up in the right order, and decide what to do about a record that does not match. This is the cheapest hour of the patrol, and the last one before the sea decides.

BRIEFING 27W

The boat sails in an hour. Every check on the rig-for-dive list exists because somebody once did not do it, and each one proves a different thing.

WHAT YOU DECIDE 13W

Prove the boat is ready before the last patrol, rather than assuming it.

13.1 What the checklist is actually asking

DAMAGE CONTROL · FORWARD EQUIPMENT & HANDLING · A ROOM · ATTEST

SCENE 37W

The rig-for-dive sheet has four signed claims. A hull valve operates; every bilge has a baseline; the drain pump builds pressure; both escape routes are clear. You have time to physically verify only two before the dive.

GUIDE 61W

Four signed claims on the rig-for-dive sheet, and time to physically verify two. Open each and read what backs it. A signature says somebody took responsibility; it does not say anybody operated the valve or walked the escape route. Hold what the evidence does not support, and spend the checks where being wrong would matter most once the boat is dived.

ASK 8W

Where should the two verification checks be spent?

BEHIND THE BUTTON 123W

Why diving changes the arithmetic. On the surface a wrong claim is an inconvenience. Dived, the same claim is a compartment nobody can leave or a valve nobody can reach. The consequence of each claim being false is what should decide the two checks, not how doubtful each looks.

What a checklist is actually asking. Not 'is this true' but 'has somebody confirmed it since the last time it mattered'. Items get signed from memory of the last dive, especially the ones that are almost always fine.

Why the escape routes are tempting to skip. They are the claim least likely to be false and the most expensive to be wrong about. That tension is the whole judgement, and it has no formula.

TAKES AS READ

a signature records a claim made earlier · a physical verification can fail even when the record is signed

VERDICT 87W

A checklist compresses work into claims, but not all claims deserve the same verification effort. The drain pump has a recent pressure test, and the bilge baseline has a current logged reading. The hull valve has only a signature from last week, while the escape route can be blocked by stores after the sheet is signed. Those two conditions can hurt immediately if they are wrong and neither has current physical backing. With only two checks, verify the critical unbacked claims rather than rechecking the best-documented items.

TAKEAWAY 20W

A signed record is evidence about what someone reported; it is not the same thing as the physical condition now.

13.2 Dampers before dive

ATMOSPHERE & LIFE SUPPORT · AUXILIARY MACHINERY & BILGE · A PERSON · SEQUENCE

SCENE 34W

The ventilation has been open to the surface all week. Before diving, the boat has to be closed up and the internal air path lined up, and the order of those two things matters.

GUIDE 70W

Ask of each card what would be much harder to correct once the hull is sealed. A missed damper and a plant that will not start are both cheap to find while the surface is still available. A baseline reading is only a baseline if it is taken before anything changes. One of these four closes the door, and everything it makes difficult it makes difficult for the whole dive.

ASK 8W

Line up the air before you need it

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

once the hull is sealed, the air aboard is the only air there is

VERDICT 84W

The internal air path should be lined up while the surface is still available. First, open the dampers that connect every compartment to the treatment plant. Next, start the scrubber and oxygen plant and prove they run. Then take handheld baseline readings, so later numbers have a trusted comparison. Only after those steps should the boat be shut off from outside air and dive. The order matters because every missed damper or failed plant becomes much harder to correct once the hull is sealed.

TAKEAWAY 14W

The air path has to be proved before the boat is sealed, not after.

13.3 What the last watch left open

FIRE & ELECTRICAL · ELECTRICAL DISTRIBUTION · A ROOM · CHOICE

SCENE 43W

The switchboard shows two breakers open from maintenance last week, but the log says one was restored. A work party is already collecting tools for that cable. Before they touch it, the chief asks you to settle which source deserves to be trusted.

GUIDE 66W

Two records disagree and a work party is collecting tools. Ask of each option what it is actually in contact with. A log records what somebody intended at the end of a watch. A lamp records what the switchboard believes about itself. Neither touches the conductor. One option would act on the disagreement rather than resolve it, which is how an inherited isolation becomes an injury.

ASK 4W

What settles the disagreement?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a record and an indicator are both claims about a circuit

VERDICT 78W

A log records what somebody intended at the end of a watch. An indicator records what the switchboard believes about itself. Neither is touching the conductor the work party will handle. An inherited isolation can therefore be wrong even when both records look plausible. A meter is the source that directly tests the cable. Its result should then be written down. The next watch inherits a measured condition instead of another claim about what somebody thought had happened.

TAKEAWAY 18W

A document and a lamp both report what somebody expects, which is a weaker thing than a reading.

13.4 Two displays, one source

NAVIGATION & POSITION • CONTROL ROOM • A ROOM • CALLBACK • CHOICE

SCENE 42W

The chart overlay and electronic repeat agree to within 100 yards. The depth sounder returns an echo after 0.120 seconds; sound speed is 1500 metres per second. The chart says 102 metres here. The last trusted position fix was 3.5 hours ago.

GUIDE 64W

Two of the readings here come from one source shown twice, so their agreement is not a second opinion. Sort the evidence by how many independent measurements stand behind it. The sounder is the only genuinely separate one. Work out what it says before choosing. A disagreement between an independent measurement and a repeated one tells you which needs checking, not which is wrong.

ASK 12W

What can you conclude from these readings before entering the banked water?

BEHIND THE BUTTON 195W

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TAKES AS READ

an echo time is a round trip • two displays that share one source can share one error

VERDICT 83W

The sounder depth is half the round-trip distance: $\frac{1}{2} \times 1500 \text{ m/s} \times 0.120 \text{ s} = 90 \text{ m}$. The chart says 102 m, so something is wrong by 12 m. The two navigation displays do not settle it because both repeat the same inertial source. Their agreement is one measurement shown twice. The sounder is independent, so the disagreement is real evidence. It does not by itself prove whether the chart or position is wrong; it proves the position needs an independent check.

TAKEAWAY 12W

Agreement only counts as confirmation when the second measurement can fail differently.

END OF THE WATCH 16W

A check is a measurement that proves a specific claim, not a box beside a word.

Refit Decision

PLAN CARD 107W

Twenty-first day, and the refit list goes off the boat in tomorrow's message window. The yard will fund one item. Haruki wants the cooling pump that has cost him two watches this patrol, Rask wants the mounts that would have prevented the noise line, Lindqvist wants a second scrubber train, and Whitfield wants to know which of them the next patrol is most likely to need. This watch you decide. The evidence is unglamorous: which failures cost real time, which are becoming more frequent, and what each fix would actually remove. Everyone in the room is arguing from what they remember, and nobody has opened the log.

BRIEFING 34W

There is money and yard time for some of what the boat wants and not all of it. Every proposal has a cost, an uncertain benefit, and an effect on the patrol that follows.

WHAT YOU DECIDE 13W

Choose what to improve before the last patrol, on evidence rather than preference.

14.1 What the boat should buy

PROPULSION & ENGINEERING · PROPULSION MACHINERY · A ROOM · ALLOCATE

SCENE 45W

The yard can complete two major jobs. Pump isolators cut radiated power by 12 dB. An independent depth sounder adds a separate transducer and power path. Faster scrubber media adds air margin; a larger drain pump raises flood-removal rate. The patrol needs two capabilities most.

GUIDE 56W

The yard can do two jobs and four are on the table. Watch the capability list rather than the equipment: it shows what the patrol plan can still do under the current pair. Every job you buy is a capability you keep and one you have given up, and the patrol needs two particular ones most.

ASK 8W

Which two refits make the patrol plan complete?

BEHIND THE BUTTON 143W

Why 12 dB is not a small number. Radiated acoustic power on a logarithmic scale: twelve decibels is a factor of about sixteen in power. That is the difference between being detectable at one range and at a quarter of it, which changes where the boat can go rather than how comfortable it is.

Why redundancy competes with performance. An independent depth sounder adds nothing on a good day and everything on the day the first one fails. Scrubber media and a larger drain pump both buy margin in a casualty. Only the patrol plan says which of those matters more.

Why two rather than three. A refit window is fixed and the yard is not yours. The choice is what makes this a decision, and the third job is not wrong — it is the one whose absence the plan can survive.

TAKES AS READ

a finite yard slot spent on one refit cannot also be spent on another · the patrol has both a stealth requirement and an old-chart navigation risk

VERDICT 89W

All four refits improve something, so asking which one is “best” hides the real decision. The patrol must reduce its pump-dominated acoustic signature. It also needs a depth measurement that can fail independently of the existing navigation chain. Faster scrubber media improves a margin that is already long. A larger drain pump helps a casualty in which stopping the source remains the binding action. With two yard slots, the complete plan is therefore pump isolation plus an independent depth sounder. Choosing either optional improvement gives up one required capability.

TAKEAWAY 17W

A resource plan is a set of questions it can answer and questions it deliberately leaves unanswered.

14.2 What 12 decibels is worth

SONAR & ACOUSTICS · SONAR ROOM · A PERSON · BALLPARK

SCENE 36W

The yard quotes a 12 dB reduction in the boat's radiated acoustic power level from new pump mounts. The wardroom has described that as "about a tenth quieter" and is weighing it against the other refits.

GUIDE 68W

Four numbers, and two are two ways of doing the same arithmetic. One counts factors of two per three decibels. One works in powers of ten. Either is legitimate, and mixing them is not. Decibels are logarithmic, so a level difference is a ratio and never a subtraction. A tenth quieter and the true factor are not the same claim, and this refit is being weighed against others.

ASK 9W

By what factor do the mounts reduce radiated power?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

decibels here are being used for an acoustic power ratio

VERDICT 76W

For a power ratio, $\Delta L = 10 \log_{10}(P_2/P_1)$. A 12 dB reduction means the original-to-new power ratio is $10^{(12/10)} \approx 15.85$. So the new radiated power is about 1/15.85, or 6.3%, of the old value. The common "3 dB is a doubling" shortcut gives nearly the same answer: 4 3 dB steps are about a factor of 16. The exact logarithmic calculation is better here because the equation is what the course is supposed to teach.

TAKEAWAY 18W

A decibel difference is a logarithmic power ratio: 12 dB corresponds to about a sixteen-fold change in power.

14.3 The value of a different failure

NAVIGATION & POSITION · CONTROL ROOM · A ROOM · CHOICE

SCENE 38W

The proposed second sounder costs £2.4m and has its own transducer and power path. The existing depth display and its repeater both use one transducer, so those two readings can agree even if that shared source is wrong.

GUIDE 60W

The existing display and its repeater share one transducer. So they can agree perfectly and be wrong together. Ask of each option whether it adds precision, adds a spare, or adds a different way to fail. Those are three different purchases. Only a measurement that fails differently can contradict a bad shared source. £2.4m is being weighed against other refits.

ASK 8W

What does the independent second sounder actually buy?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

two displays sharing one sensor share that sensor's failures

VERDICT 85W

Two displays fed by one transducer are not two independent measurements. They can agree perfectly while sharing the same sensor error, power fault or calibration problem. A second sounder with its own transducer and power path adds a genuinely different failure mode. It does not make either instrument more precise. It also does not guarantee the chart is right. It gives command a second physical measurement that can disagree for a meaningful reason. That is how a bad shared source gets detected instead of repeated.

TAKEAWAY 15W

Redundancy is valuable when the backup has a different failure path, not merely another display.

14.4 What 90 metres puts behind the valve

DAMAGE CONTROL · FORWARD EQUIPMENT & HANDLING · A ROOM · CALLBACK · BALLPARK

SCENE 40W

The suspected hull connection is 90 metres below the surface. Ferro writes seawater density as 1025 kilograms per cubic metre and g as 9.81 metres per second squared. He wants the gauge pressure on the sea side of the fitting.

GUIDE 78W

Five numbers, and they are not all about the same thing. Two describe the bilge: how fast the level rose, and what a centimetre of it holds. One is what the drain pump can shift at this head. One is the depth, and one is where the level started. Ask of each whether it describes the water coming in or the water going out. A rate the pump cannot beat is a different situation from one it can.

ASK 12W

About how much gauge pressure is the sea applying at 90 metres?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

gauge pressure counts the pressure added by the water column above the surface value · buoyancy and Archimedes — taken as read

VERDICT 78W

Hydrostatic pressure is $p = \rho gh$. Use seawater density 1025 kg/m³, $g = 9.81$ m/s², and $h = 90$ m. The gauge pressure is about 905,000 pascals, or 0.91 megapascals. That is roughly nine atmospheres above surface pressure. Add the atmosphere already present, and the absolute pressure outside the hull is about ten atmospheres. This is why a small opening at depth can drive a large inflow and why the hull isolation matters more than adding another pump.

TAKEAWAY 9W

Hydrostatic gauge pressure rises in direct proportion to depth.

END OF THE WATCH 14W

An upgrade is worth what it changes about the decisions you will actually face.

Deep Watch

BEFORE THE WATCH — TAG

Catch Rask before she goes into the control room

Rask, the sonar supervisor, has the sound survey from the last patrol, which is the only record of what this hull sounded like when it was quiet. Once she is in the control room and the brief has started she is unavailable for two hours, and the whole afternoon's work is a comparison against a survey you do not yet have a copy of.

PLAN CARD 118W

Twenty-second day, the last full watch of the patrol, and everything is degraded a little. A contact Rask has not resolved. A plot Sowande has not been able to fix since the forenoon. A machine Haruki has not looked at since Tuesday. Whitfield is in the wardroom with the corpsman and a fractured wrist of his own, so the officer you would have handed half of this to is not available. This watch you decide the order everything gets worked in, and you do it while it is happening. Nobody will tell you which problem is the important one. Getting that ranking right is the job, and it is the only part of it that cannot be delegated.

BRIEFING 33W

An uncertain contact, a plot four hours old, water in the forward bilge, a scrubber train down and a bearing running warm. None of them is an emergency. Together they are the patrol.

WHAT YOU DECIDE 16W

Run the whole boat at once, with everything degraded a little and nothing decided for you.

15.1 Five small problems

NAVIGATION & POSITION · CONTROL ROOM · A ROOM · CHOICE

SCENE 43W

Nothing on the board is red. The plot is four hours old, the contact is faint, the forward bilge is wet, one scrubber train is down, and a bearing is warm. The constrained passage is still ahead, and Whitfield asks what changes first.

GUIDE 69W

Nothing on the board is red, so severity will not rank these. Ask of each what is growing: uncertainty, consequence, or neither. A faint contact needs time before it can be read at all. A wet bilge matters if the level is moving. One scrubber train down is a deadline rather than an emergency. Constrained water is ahead, and one of these gets worse simply by the boat continuing.

ASK 9W

What is the first thing that has to change?

BEHIND THE BUTTON 195W

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TAKES AS READ

trend and time-to-limit matter more than whether a reading merely looks abnormal

VERDICT 86W

Five small problems compete for attention, but they do not age at the same rate. The old plot gets more dangerous as constrained water approaches. The faint contact needs time to develop geometry. A wet bilge matters if its level is changing. One scrubber train down is a deadline if carbon dioxide is rising. A warm bearing matters if its cooling margin is shrinking. The first move should change the condition whose uncertainty or consequence is growing fastest, then give the others owners and return thresholds.

TAKEAWAY 18W

A set of small problems can be worse than a big one, because none of them demands attention.

15.2 Everything you are not doing yourself

DAMAGE CONTROL · FORWARD EQUIPMENT & HANDLING · A PERSON · PROTOCOL

SCENE 43W

You keep the degraded plot yourself while four other problems remain open. Whitfield stands beside the watch bill with names ready, but nobody is assigned yet. This is the last watch of the patrol, and the handoff has to work without you hovering.

GUIDE 71W

Four open problems and four assignments, and the pairing is about what each person is meant to notice. Ask of each item what the useful report would be: a change, a rate, an hour, or a cost. A number logged with no threshold is a number nobody acts on. This is the last watch of the patrol, so an assignment that still needs you standing over it has not been made.

ASK 9W

Match each item to the assignment that fits it.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

an instruction has to be actable by the person who receives it

VERDICT 86W

A useful assignment lets the other person act without guessing what you meant. It names an owner, a first action and a condition for reporting back. For the bilge, log the level and report a change. For the contact, hold it long enough to obtain bearing rate. For the scrubber, restore it or name the deadline. For the warm bearing, restore cooling and state the acoustic cost. "Watch it" sounds responsible, but it gives no one a decision point and leaves every problem attached to you.

TAKEAWAY 17W

An assignment that names a threshold can be acted on alone; one that says watch it cannot.

15.3 Why the level pulses three times a second

SONAR & ACOUSTICS · SONAR ROOM · A ROOM · BALLPARK

SCENE 38W

2 narrowband components sit at 147 and 150 hertz on the same bearing. In seawater the sound speed is about 1500 metres per second. The audio level swells and fades regularly while both spectral lines themselves remain steady.

GUIDE 67W

Three numbers, and one of them is the sound speed, which the pulsing rate does not need at all. It belongs to the wavelengths rather than to the beat. Ask of each number whether the swelling depends on it. Both spectral lines are steady, so nothing is changing frequency. What is changing is how the two patterns line up, and that comes from the gap between them.

ASK 11W

How many amplitude swells per second should the two tones produce?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

waves add when they arrive together · sound speed is approximately the same for these two nearby frequencies · sound as a wave: frequency, wavelength, speed — taken as read

VERDICT 82W

The beat frequency is $f_{\text{beat}} = |f_1 - f_2|$, so $150 - 147 = 3$ Hz: three amplitude swells each second. The wave relation $v = f\lambda$ also gives different wavelengths in the same water: about $1500/147 = 10.20$ m and $1500/150 = 10.00$ m. Because the two wave patterns advance with slightly different phase rates, they repeatedly line up and oppose each other. The pulsing is therefore interference between close-frequency components, not a third spectral line or a changing shaft speed.

TAKEAWAY 14W

Two nearby frequencies beat at their difference frequency because their relative phase cycles steadily.

15.4 The one shaft speed that rings

PROPULSION & ENGINEERING · PROPULSION MACHINERY · A ROOM · CALLBACK · CHOICE

SCENE 30W

At 92 revolutions the after compartment rings loudly enough to hear through a bulkhead. At 85 and at 100 it does not. Nothing else about the plant changes between them.

GUIDE 74W

All four options are real machinery faults and three of them get steadily worse with speed. Ask of each whether it would come and go at one speed and nowhere else. Imbalance grows, bearings degrade, cavitation starts and stays. Only something that depends on matching a frequency can be loud at ninety-two revolutions and quiet at eighty-five and a hundred. That distinction decides whether the fix is a repair or an avoided speed band.

ASK 13W

The hull rings at one shaft speed and nowhere else. Why that one?

BEHIND THE BUTTON 195W

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TAKES AS READ

every structure has frequencies at which it prefers to vibrate · sound as a wave: frequency, wavelength, speed — taken as read

VERDICT 89W

A structure has natural vibration modes. If a repeating force lands close to one mode, each cycle adds energy in step with the motion and the vibration grows. Blade rate is shaft revolutions per second multiplied by the number of blades. One shaft speed can therefore put that forcing near a hull or mount resonance while nearby speeds do not. That is why the practical fix can be an avoided speed band or a

changed mount stiffness. The water-wave relation $v = f\lambda$ is not the governing equation here.

TAKEAWAY 21W

A structure has natural vibration frequencies, and forcing near one of them can make a small input produce a large response.

END OF THE WATCH 14W

Command is choosing which uncertainty you can live with, and saying so out loud.

THE CLOSING CARD

The boat surfaced at dawn on the twenty-third day, into weather nobody had seen for three weeks. The fractured wrist is in a cast and signed fit for the transit. The contact was resolved, the plot was fixed, and the machine that had been left since Tuesday was opened up and found in time.

Nothing was lost on this patrol that could not be repaired. The whole crew went up the ladder into daylight, which is the only measure of a watch that matters.

Card text only — no options, boards or answers. 13,188 words of card prose across 61 stops. Generated from themes/deepwatch by tools/make-cardbook.mjs.